ENTERPRISE AIOPS PLATFORM • ARCHITECTURE SPECIFICATION

# Ingestion Architecture & Streaming Pipelines

**Target Platform:** Google Cloud Platform (GCP)

**System of Record:** ServiceNow ITSM

**Version:** 2.0 (Production Blueprint)

**Generated:** August 2026

## Ingestion Layer & Streaming Pipelines Architecture

This document outlines the **Enterprise AIOps Ingestion Engine**, which captures high-throughput telemetry from multi-cloud and edge observability sources, standardizes and sanitizes it in real time, and routes it to downstream analytics and incident response systems natively on **Google Cloud Platform (GCP)**.

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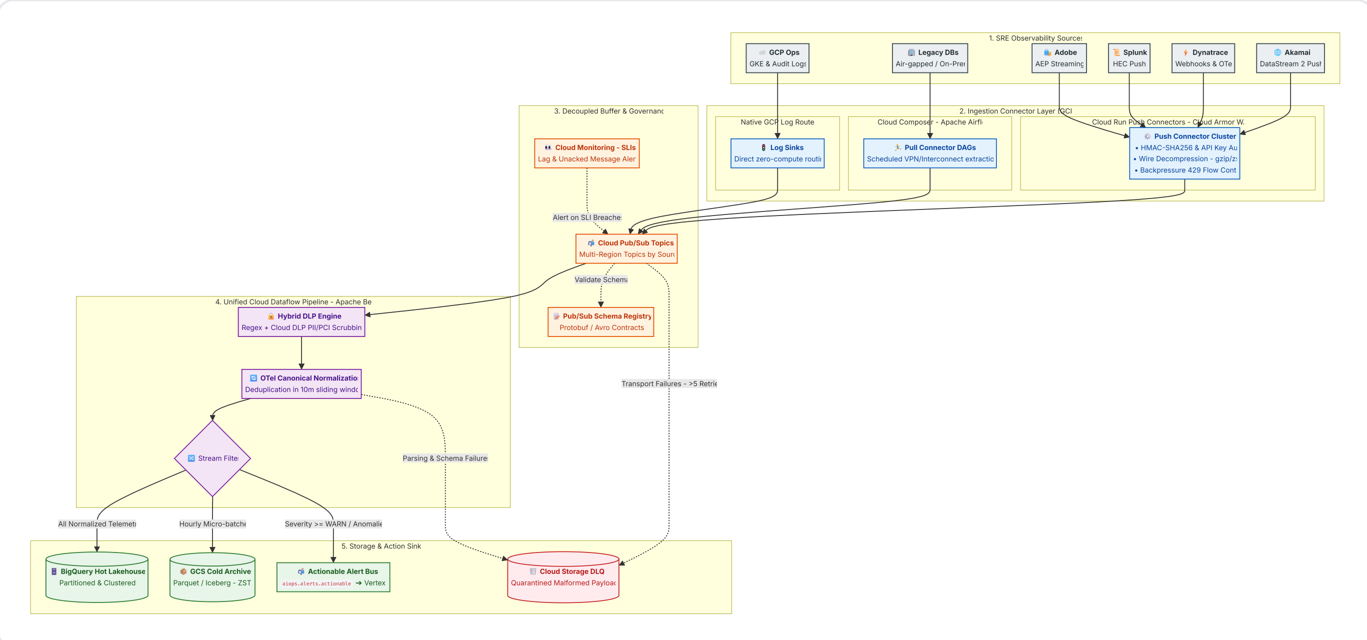
### 1. High-Level Ingestion Architecture

The ingestion architecture follows a decoupled, connector-based pattern capable of handling 150,000 to 500,000+ Events Per Second (EPS) with sub-second latency and zero data loss.

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### 2. The Ingestion Connector Layer

Telemetry sources exhibit different networking capabilities and security constraints. We utilize three distinct connector patterns to ingest telemetry securely into GCP:

### 2.1 Push Connectors (Modern SaaS & Edge Ingress)

Used by sources capable of real-time outbound streaming (**Akamai DataStream 2**, **Dynatrace Webhooks**, **Splunk HEC**, **Adobe Analytics AEP**).

- **Ingress Security**: Traffic passes through **Cloud Armor WAF** with strict IP allowlisting (filtering for vendor CIDR ranges) and volumetric DDoS rate limiting.
- **Compute Layer**: Auto-scaling, stateless **Cloud Run** container services deployed across multiple zones.
- **Authentication & Webhook Validation**:
  - **API Keys & Tokens**: Verified dynamically against **GCP Secret Manager** with in-memory caching.
  - **HMAC-SHA256 Signatures**: Validates webhook payload authenticity using vendor signature headers (e.g., `X-Akamai-Signature`, `X-Adobe-Signature`) to prevent request spoofing.
- **Wire Compression**: Enforces `Content-Encoding: gzip` / `zstd` on inbound streams, reducing cross-cloud data transfer volume and egress bandwidth costs by up to 70%.
- **Backpressure & Flow Control**: If downstream Pub/Sub publishing encounters transient backpressure, Cloud Run returns `HTTP 429 (Too Many Requests)` or `HTTP 503` with a `Retry-After` header, prompting upstream SaaS forwarders to buffer and retry exponentially.

### 2.2 Pull Connectors (Legacy & Air-Gapped Systems)

Used by on-premise relational databases, mainframes, or network appliances that cannot push outbound traffic.

- **Orchestration**: **Cloud Composer (Managed Apache Airflow)** schedules and executes extraction DAGs.
- **Networking**: Traffic travels over private, encrypted tunnels via **Cloud VPN** or **Cloud Interconnect**.
- **Extraction & Retries**: Airflow operators extract delta records, convert them to canonical JSON/Avro, and publish to Cloud Pub/Sub with automatic exponential backoff retries.

### 2.3 Native Connectors (GCP Operations Suite)

Used natively by Google Cloud Platform workloads (**Google Kubernetes Engine (GKE)**, **Cloud Run**, **Cloud Audit Logs**).

- **Mechanism**: Leverages **Cloud Logging Log Router Sinks** pointing directly to Cloud Pub/Sub topics.
- **Advantage**: Zero-compute, zero-maintenance ingestion with sub-second delivery latency and zero egress cost.

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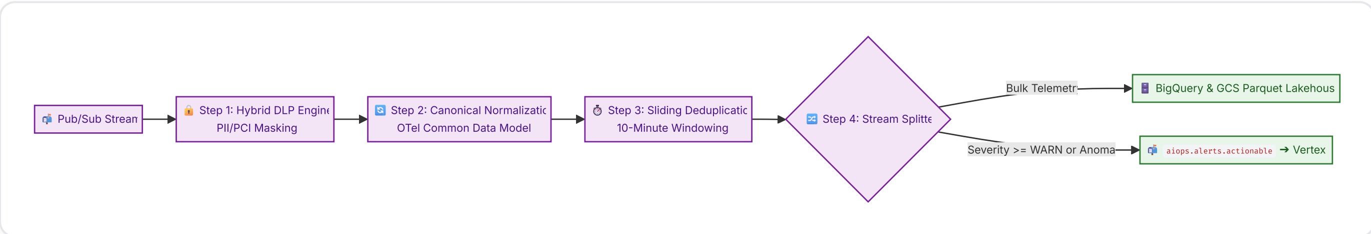
## 3. Decoupled Buffer & Stream Processing

### 3.1 Cloud Pub/Sub (The Multi-Region Event Bus)

- **Topic Isolation**: Dedicated, isolated topics per source (e.g., `telemetry.akamai.raw`, `telemetry.dynatrace.raw`, `telemetry.splunk.raw`) prevent single-source traffic storms (such as a DDoS attack logged by Akamai) from starving other telemetry streams.
- **Multi-Region Durability**: Topics are configured with multi-region message routing policies to ensure continuous ingestion availability during single-region GCP outages.

### 3.2 Cloud Dataflow (Unified Apache Beam Streaming Pipeline)

A unified, horizontally auto-scaling streaming pipeline processes incoming telemetry in four sequential stages:




1. **Hybrid DLP Engine**: Fast, in-memory regex tokenizers redact high-risk payment card numbers (PCI-DSS) and social security numbers. Unstructured log bodies are asynchronously scrubbed using the **Cloud Data Loss Prevention (DLP) API**.
2. **Canonical Normalization**: Standardizes disparate formats (Akamai JSON, Splunk HEC, Dynatrace PurePath, Adobe clickstream) into an **OpenTelemetry-aligned canonical schema** (`aiops_lakehouse.telemetry_canonical`).
3. **Sliding Window Deduplication**: Removes duplicate events within a 10-minute sliding window based on `(source_tool, source_event_id, timestamp)`.
4. **Intelligent Stream Splitting**:
   - **Bulk Analytical Stream**: Routes 100% of clean telemetry to BigQuery (Hot Tier) and GCS Parquet (Cold Tier).
   - **Actionable Alert Stream**: High-severity anomalies and warning events (`severity ≥ WARN` or Davis AI problems) are extracted and published to the high-priority Pub/Sub topic `aiops.alerts.actionable` for consumption by the **Vertex AI Semantic Router**.

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## 4. Reliability, Robustness & Governance Patterns

### 4.1 Schema Management & Dynamic Evolution

- **Pub/Sub Schema Registry**: Enforces Protobuf and Avro schema contracts at topic ingress.
- **Schema Drift Protection**: Incoming records that introduce backward-compatible fields are parsed dynamically into `raw_attributes JSON`, preventing Dataflow worker crashes while preserving full fidelity.

### 4.2 Multi-Tier Dead-Letter Queues (DLQs)

To prevent unparseable or poisoned payloads from blocking the pipeline:

1. **Transport-Level DLQ (Pub/Sub)**: If Dataflow fails to acknowledge a message after 5 delivery attempts, Pub/Sub diverts it to a native dead-letter topic ( `telemetry.*.dlq` ).
2. **Application-Level DLQ (Dataflow Side-Output)**: Payloads that parse structurally but fail semantic validation (e.g., corrupted timestamps) are emitted via Beam side-outputs.
3. **Storage & Replay**: Both DLQ streams persist to a date-partitioned **Cloud Storage DLQ Bucket** ( `gs://aiops-dlq-bucket/YYYY/MM/DD/` ) for offline inspection, alerting, and automated replay.

### 4.3 Ingestion Service Level Indicators (SLIs) & Alerting

The platform team continuously monitors pipeline health in **Cloud Monitoring**:

| Metric Name | Target SLI | Alert Severity | Action upon Breach |
|---|---|---|---|
| `oldest_unacked_message_age` | $< 5$ minutes | **P1 (Critical)** | Pages Data Platform SRE via ServiceNow webhook |
| `dataflow/system_lag` | $< 120$ seconds | **P2 (Major)** | Triggers automated worker auto-scaling |
| `dataflow/watermark_age` | $< 180$ seconds | **P2 (Major)** | Alerts on upstream source timestamp drift |
| `dlq_message_count` | $< 100$ msgs/hour | **P3 (Warning)** | Flags upstream schema violation for review |

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## 5. Related Architecture Specifications

- SRE Observability Fleet - Source Telemetry Matrix
- Data Contracts & Canonical Schemas
- Ingestion Best Practices Guide
- Data Processing & AI Feature Store